The exponential wall: a finite-size crossover criterion for shadow-based moment estimation, with a hardware case study

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Abstract

Classical shadows estimate nonlinear functions of a quantum state, such as the moments $\mathrm{Tr}(\rho^k)$, from randomized single-copy measurements. The finite-sample variance of the resulting U-statistic estimators splits into terms of different budget order — a Hoeffding decomposition [17, 18] stated for shadows in [1] — whose two decay regimes are known [19, 21]. Those analyses bound the state-dependent projection variances, and separate bounds on them do not constrain their ratio. We evaluate them in closed form for depolarized Haar-random pure states, locating the budget $M^* = \zeta_2/(2\zeta_1)$ at which the effective error exponent crosses over. It grows exponentially, with asymptotic base 28/5 following from the ensemble-averaged projection variances and approached from below by the finite-size fits over the sizes we reach ($n = 2$ to 9).

That evaluation turns the regime structure into a plug-in criterion, with nothing fitted to the crossover data, for when a collective two-copy measurement attains a lower RMSE at equal copy budget than single-copy shadows. Across the 83 simulated cells that resolve a crossover, spanning four state ensembles with three held out from the derivation, it predicts the crossover size to within one qubit in 82, and in 95.9% of all 123 swept cells once non-crossing cells are scored as predictions; simulating readout on both routes leaves those sizes a conservative upper bound. The cells are simulation, and for $k > 2$ the collective side is modelled as a bounded ± 1 outcome carrying none of the gates its circuit would require.

We then confront the prediction with what the platform reports as a 108-qubit superconducting processor, with predictions committed before submission. Both failed: a Bell-state purity locked at 0.89 measured 0.72, and a single-copy anchor locked at 1.50 measured 1.34, with disjoint intervals. Byte-identical circuits also varied substantially between sessions. We do not demonstrate the crossover on hardware, and the failure is informative: static, single-session characterization did not predict collective-measurement performance on the platform and execution path we tested.

1. INTRODUCTION

At ten qubits, on the noisy-pure states we benchmark and at a fixed measurement budget, the standard single-copy method for estimating the purity of a quantum state returns an error fifteen times larger than the quantity it is estimating. The estimator is still unbiased; it is the spread that has become useless.

That matters because the quantity is not exotic. The moments $\mathrm{Tr}(\rho^k)$ are the entry point to much of what one wants to know about an unknown state: the second is the purity, their logarithms give the Rényi entropies, moments of the partial transpose furnish PPT entanglement criteria and bounds on negativity, and purity enters error-mitigation protocols such as virtual distillation. Any experiment characterizing the state its device actually prepared needs these numbers.

Two routes exist. The single-copy route measures one copy at a time in randomized local bases, building classical shadows [1], and forms an unbiased U-statistic from the snapshots; it uses no entangling gates. The collective route holds k copies simultaneously and measures the cyclic-permutation operator, whose expectation is exactly $\mathrm{Tr}(\rho^k)$; it has $O(1)$ variance but requires entangling operations across copies.

The asymptotic separation is settled. Single-copy strategies require exponentially more samples than collective ones for a family of learning tasks [2, 3], and for protocols restricted to an identical single-copy projection-valued measurement the lower bound for purity estimation is $\Omega(\max\{1/\varepsilon^2, 2^{n/2}/\varepsilon\})$ [4]. What is not settled is the question an experimentalist faces: on this device, at this noise level, at this system size, which route has the lower RMSE at equal state-copy budget? Equal copy budget is not equal cost — the routes differ in simultaneous qubit count, circuit count, depth, entangling gates, routing, and calibration — and the collective route buys variance reduction by paying in noise-induced bias. Whether that trade is worth making depends on numbers that have not been computed.

The gap is visible in recent work. A May 2026 study of quantum machine learning advantage on tens of noisy qubits [5] declined to give its measure-first protocol multi-copy access, citing both a result specific to its learning task and the resource demands of multi-copy entangled measurements. That task is not moment estimation, but the caution about resource cost is a premise we can test quantitatively. On the theory side, [6] shows that the exponential two-copy advantage for purity *testing* collapses under local depolarizing noise;

their task is testing and ours is estimation, a distinction we take up in Section 7. The caution is reasonable. What has not been available is a prediction of the size at which it stops applying.

That the finite-sample variance splits into projection terms of different budget order is the standard Hoeffding identity [17, 18], stated for shadows in [1], and that the resulting decay rate changes with the budget is established: Elben et al. [19] state both regimes and plot them, and Aasen and Gärttner [21] track the exponent migrating between them under detector noise. What those analyses do not do is locate where the regimes meet. Bounding ζ_1 and ζ_2 separately constrains their ratio not at all, and the exact second moment that would fix it is applied in the literature to worst-case sample complexity rather than to a state.

Thesis. We evaluate the state-dependent projection variances for depolarized Haar-random pure states in closed form and feed them into the exact variance formula. That fixes the crossover budget $M^* = \zeta_2/(2\zeta_1)$, which grows exponentially in n, and, set against two exact bias laws for the collective route, yields a prediction — with nothing fitted to the crossover data — of the size at which a collective two-copy measurement attains the lower RMSE at equal copy budget. The same evaluation runs at $k = 3$ and 4, where prior treatments bound the higher-moment variance, Elben et al.'s Appendix D.2 among them, and where the two-term truncation no longer suffices. A hardware comparison of the two routes has been reported [20]; what we add is the predictive law and a test of it locked in advance. That test failed, and what it established is the shape of the obstacle: the dominant error was readout rather than the entangling overhead the crossover argument prices, and device parameters characterized in earlier sessions did not describe the device we ran on.

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2. SETTING AND ESTIMATORS

2.1. Local random-unitary classical shadows

Let ρ be an unknown n-qubit state. A single snapshot is produced by drawing an independent Haar-random single-qubit unitary U_q for each qubit, applying $\bigotimes_q U_q$, measuring in the computational basis to obtain a bitstring b, and forming

$$G = \bigotimes_{q=1}^{n} \left(3\, U_q^\dagger |b_q\rangle\langle b_q| U_q - \mathbb{I}\right).$$

The snapshot is an unbiased estimator of the state, $\mathbb{E}[G] = \rho$. A budget of M snapshots consumes M copies of ρ.

2.2. The estimator must be the exact U-statistic

The k-th moment is estimated by the U-statistic over distinct k-tuples of snapshots,

$$U_M = \binom{M}{k}^{-1} \sum_{i_1 < \cdots < i_k} h(G_{i_1}, \ldots, G_{i_k}), \qquad h(g_1, \ldots, g_k) = \frac{1}{k!} \sum_{\pi \in S_k} \mathrm{Re}\,\mathrm{Tr}\big(g_{\pi(1)} \cdots g_{\pi(k)}\big),$$

which is exactly unbiased: $\mathbb{E}[U_M] = \mathrm{Tr}(\rho^k)$.

The tuples are formed exhaustively rather than sampled. Forming them from already-collected snapshots is classical post-processing and consumes no additional copies, so subsampling saves nothing physical while inflating the variance — in our sweeps it inflated the single-copy RMSE by a factor of about five at $n = 2$ and more than fiftyfold at $n = 4$, enough to reverse the benchmark's conclusion. Any single-copy-versus-collective comparison must use the exact estimator on the single-copy side.

For $k = 2$ the full U-statistic has an exact closed form, and the two ways of evaluating it trade M against n. The power-sum reduction $\mathrm{Tr}(S^2) - \sum_i \mathrm{Tr}(G_i^2)$, with $S = \sum_i G_i$, is linear in M but builds dense $2^n \times 2^n$ matrices at $O(M4^n)$; we instead use the per-qubit factorization $\mathrm{Tr}(G_i G_j) = \prod_q \mathrm{Tr}(G_i^q G_j^q)$, forming an $M \times M$ pairwise-trace matrix in $O(M^2 n)$ time. For $k = 4$ no complete power-sum reduction exists, because the snapshots do not commute; Appendix A gives an exact construction by Möbius inversion over the 15 set partitions of the four cyclic slots, with the alternating partition evaluated by tensor contraction, matching brute-force enumeration to $\lesssim 10^{-13}$.

2.3. The collective route

Let C_k be the cyclic permutation operator on k registers of dimension $d = 2^n$. It satisfies

$$\mathrm{Tr}\big(C_k\, \rho^{\otimes k}\big) = \mathrm{Tr}(\rho^k), \qquad \mathrm{Tr}(C_k) = d,$$

the second identity because the cyclic permutation has a single cycle, and it is what makes the depolarizing bias law of Section 4.1 come out linear in the noise rate.

We use the destructive SWAP test rather than the ancilla-based controlled-SWAP construction, which decomposes into many two-qubit gates and would be dominated by gate noise on current hardware. Two copies are prepared on $2n$ qubits; for each pair $(i, i+n)$ a CNOT is applied with control i and target $i+n$, followed by a Hadamard on qubit i, and all $2n$ qubits are measured. The purity is recovered by the parity rule

$$\mathrm{Tr}(\rho^2) = \sum_{\text{bitstrings}} (-1)^{\#\{\, i\,:\, a_i = b_i = 1\,\}}\, P(\text{bitstring}),$$

where a_i, b_i are the outcomes on the two copies of qubit i. The circuit uses exactly n two-qubit gates, requires no ancilla, and has depth two after state preparation; we verified the sign rule against exact simulation to 10^{-16} (Appendix B). That gate count is why the entangling overhead is affordable on real hardware (Section 6).

This construction is specific to $k = 2$, where C_2 is Hermitian with eigenvalues ± 1 so its expectation is directly a two-valued measurement. For $k \geq 3$ the cyclic operator is not Hermitian and no analogous destructive parity test exists, so we take the standard ancilla-based Hadamard test: an ancilla in $|+\rangle$, a controlled-C_k on the k copies, and an X-basis measurement of the ancilla, whose ± 1 outcome has expectation $\mathrm{Re}\,\mathrm{Tr}(C_k \rho^{\otimes k}) = \mathrm{Tr}(\rho^k)$. Its shot variance is $1 - \mu^2$ exactly as at $k = 2$, which is the model used for all k in Section 5. The cost is not comparable: transpiled to the tested coupling map, the controlled cyclic shift on kn qubits uses about ten times the two-qubit gate count of the $k = 2$ destructive test at $k = 3$ and about sixteen at $k = 4$, and needs an ancilla and routing it does not. Our $k \geq 3$ results are simulated, and the gate counts in Section 6.2 are for $k = 2$ only.

2.4. Copy-fair accounting

A single-copy protocol at budget M consumes M copies of ρ. A k-copy collective test at budget M consumes $\lfloor M/k \rfloor$ measurements, each using k copies (the swept budgets are not all multiples of k). All comparisons in this paper hold the total number of state copies fixed.

2.5. The state ensemble is not a free choice

The variance of shadow-based purity estimation scales with the purity of the state being measured, a fact we derive exactly in Section 3.3. The consequence is a trap, though not the one it first appears to be. A random Ginibre state under heavy depolarizing noise has purity approaching 2^{-n}, and a quantity that is approximately zero is easy to estimate at fixed absolute error — though not at fixed relative precision, nor for testing, where the gap to be resolved shrinks too (Section 7). What does not follow is that the estimator is well behaved there.

Low purity does not remove the exponential cost; it concentrates it. At $\rho = \mathbb{I}/2^n$ every snapshot gives $\mathrm{Tr}(G\rho) = 2^{-n}$ deterministically, so $\zeta_1 = 0$ exactly and $M^* = \zeta_2/(2\zeta_1)$ diverges, putting the estimator in the higher-order regime at every budget. The kernel term does not vanish with it: the single-qubit second moment $\mathbb{E}[\mathrm{Tr}(G_1G_2)^2]$ is 7 for a maximally mixed qubit, so $\zeta_2 = 7^n - 4^{-n}$, and the variance $2\zeta_2/M(M-1)$ gives an error of order $7^{n/2}/M$ — exponential in n, against a target itself shrinking as 2^{-n}.

The exponential is therefore a property of the estimator, not of high-purity states. Its base is set by ζ_2, which does not depend on purity at all but on how the state's Pauli weight is distributed: it is exactly 7 per qubit both at the maximally mixed state and for any spread Pauli spectrum, and larger for stabilizer-structured spectra (Section 3.5). Purity governs ζ_1, and so M^* and the exponent, not the exponential itself. The ensemble nonetheless decides what the benchmark measures: at $\rho = \mathbb{I}/2^n$ the collective bias also vanishes identically for every *unital* channel, since $\mathrm{Tr}(\rho^k)$ then sits exactly on the depolarizing floor $2^{n(1-k)}$, so both routes degenerate at once and the comparison stops being informative. The restriction matters: amplitude damping is not unital, and carries $\mathbb{I}/2$ to a state of purity $(1+\gamma^2)/2$.

We therefore benchmark on noisy-pure states, a prepared pure state under modest depolarizing noise,

$$\rho = (1-q)|\psi\rangle\langle\psi| + q\,\mathbb{I}/2^n, \qquad |\psi\rangle \sim \text{Haar}, \qquad q \in [0.05,\, 0.3],$$

whose purity is stable across the sizes tested but set by q, approaching $(1-q)^2$ at large n: about 0.90 at $q = 0.05$, 0.81 at the representative $q = 0.1$, and 0.49 at $q = 0.3$. These states keep the problem hard and meaningful at once — the moment is of order one, so an absolute error is a statement about something, and the collective route has a bias to pay, so the routes are in competition. Choosing the wrong ensemble does not make the estimator cheap; it makes the comparison vacuous.

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3. THE SINGLE-COPY VARIANCE LAW

3.1. The Hoeffding decomposition gives the variance exactly

For a symmetric kernel h of order k, define the c-th order projection and its variance,

$$h_c(g_1, \ldots, g_c) = \mathbb{E}_{G_{c+1},\ldots,G_k}[h(g_1, \ldots, g_c, G_{c+1}, \ldots, G_k)]\,, \qquad \zeta_c = \mathrm{Var}[h_c(G_1, \ldots, G_c)]\,.$$

The variance of the U-statistic is then exactly

$$\boxed{\mathrm{Var}(U_M) = \binom{M}{k}^{-1} \sum_{c=1}^{k} \binom{k}{c}\binom{M-k}{k-c}\zeta_c}$$

This is the standard Hoeffding decomposition for U-statistics [17, 18], and we claim no novelty for it; what is new here is the closed-form evaluation of the second-order ζ_c and the threshold their ratio fixes (Section 3.5).

For $k = 2$ this reduces to

$$\mathrm{Var}(U_M) = \frac{4(M-2)\,\zeta_1 + 2\,\zeta_2}{M(M-1)}, \qquad \zeta_1 = \mathrm{Var}[\mathrm{Tr}(G\rho)]\,, \quad \zeta_2 = \mathrm{Var}[\mathrm{Tr}(G_i G_j)]\,.$$

We verified the general formula against brute-force Monte Carlo of the exact estimator at $k = 3$ and $k = 4$, on noisy-pure, GHZ, and low-rank states, with all ratios within 3%, and confirmed that it reduces to the $k = 2$ form at a ratio of 1.000000. The law is exact, not asymptotic, and it holds at every budget.

Because the k-argument kernel is multilinear in the snapshots, every projection variance depends on the shadow ensemble only through its second-moment matrix; local Haar single-qubit shadows and local Pauli-basis shadows have identical second-moment matrices, so all results of this section hold verbatim for either.

One subtlety is easy to get wrong and can pass unnoticed. For $k = 2$ the second-order projection *is* the kernel, so ζ_2 is the kernel variance. For $k \geq 3$ this is false: ζ_2 is the two-argument projection, and the kernel variance is ζ_k. Substituting one for the other is wrong by roughly a factor of seven and corrupts any variance estimate at higher moments while appearing to work at $k = 2$. The resulting approximation breaks down at $k \geq 3$, for reasons that can look like physics but are not.

3.2. Two terms compete, and their ratio is a budget threshold

The exact variance contains two competing terms with different budget dependence. At large M the linear term dominates,

$$\mathrm{Var}(U_M) \approx \frac{4\zeta_1}{M} \quad \Longrightarrow \quad \mathrm{RMSE} \propto M^{-1/2}, \qquad \alpha = \tfrac{1}{2},$$

which is the familiar statistical scaling that a fixed-exponent bound predicts. At small M the higher-order term dominates,

$$\mathrm{Var}(U_M) \approx \frac{2\zeta_2}{M^2} \quad \Longrightarrow \quad \mathrm{RMSE} \propto M^{-1}, \qquad \alpha = 1.$$

The two asymptotic terms are equal at

$$\boxed{M^* = \frac{\zeta_2}{2\zeta_1}}$$

This is the balance point of the two large-M forms above; the exact numerator terms balance at a point differing from it by an additive constant, immaterial once M^* is large. Nothing that follows depends on the choice, because the predicted exponents are computed from the exact variance formula rather than from M^*. This threshold is the object the rest of the section is about. Which side of M^* a given experiment sits on determines the character of its scaling, and the projection variances that set M^* are state-dependent. To see what that dependence is, we compute them exactly for one qubit.

3.3. The single-qubit second moment, exactly

For a single qubit with density matrix r — throughout this subsection we write r for a single-qubit state, reserving ρ for the full n-qubit state, because the identity below holds only at one qubit — with Bloch length t and purity $p = (1 + t^2)/2$,

$$\boxed{\mathbb{E}\big[\mathrm{Tr}(Gr)^2\big] = \tfrac{1}{4} + \tfrac{5}{4}t^2 = \tfrac{5}{2}p - 1}$$

and the single-qubit first projection variance follows as

$$\zeta_1 = \tfrac{3}{4}t^2 - \tfrac{1}{4}t^4.$$

We verified this numerically across the full Bloch range. The identity is exact, and two consequences follow from it.

The second moment grows by a factor of six from a maximally mixed qubit, where it equals 1/4, to a pure qubit, where it equals 3/2. Compounded over n qubits, that factor is the origin of the exponential cost, and it is state-dependent rather than universal. The compounding picture is a heuristic for the factorized case: it does not carry over to arbitrary entangled or correlated states, which is what Section 3.4 shows when the weight-only ansatz fails at $n \geq 2$.

The identity also passes the boundary check. At $t = 0$ it gives $\zeta_1 = 0$, which is correct: for a maximally mixed qubit, $\mathrm{Tr}(Gr) = 1/2$ deterministically, with no variance to speak of.

3.4. The weight-only quadratic ansatz fails already at one qubit

The single-qubit result invites a generalization. If the second moment carries a factor of 3 per qubit in the support of a Pauli string, then ζ_1 ought to decompose over Pauli strings with a coefficient depending only on the Pauli weight,

$$\zeta_1 \stackrel{?}{=} \sum_P c_{|P|} \langle P \rangle^2.$$

This already fails at a single qubit, and provably so. With universal weight coefficients the most general such form for one qubit is $c_0\langle I\rangle^2 + c_1(\langle X\rangle^2 + \langle Y\rangle^2 + \langle Z\rangle^2) = c_0 + c_1 t^2$, an affine function of t^2, because the three squared Pauli expectations sum to the squared Bloch length t^2 and carry no further invariant. The exact single-qubit projection variance of Section 3.3 is $\zeta_1 = \frac{3}{4}t^2 - \frac{1}{4}t^4$, whose quartic term has coefficient $-\frac{1}{4}$ whatever c_0 and c_1 are chosen: no weight-only quadratic form reproduces it. The obstruction is a quartic dependence on the state that a form quadratic in the Pauli expectations cannot carry, and it is present before any question of entanglement or correlation arises.

The insufficiency persists and grows with size. Tested against a diverse family of 19 states at $n = 2$, varying depolarizing rate, rank, and structure, the best-fit weight-only ansatz leaves a residual of up to 12.3%. A weight-only quadratic form is insufficient, and Section 3.5 identifies exactly what it omits. The weight-only form is the diagonal of an exact cubic identity already available in [1], and the residual it leaves is that identity's off-diagonal sum.

The ansatz nonetheless *appears* to hold. Tested only within a narrow single-parameter family, Haar-pure states at a fixed depolarizing rate of $q = 0.1$, it fits to 0.1%, because that family spans a low-dimensional invariant subspace. A closed form validated on one ensemble is not a closed form. At second order the projection variances follow in closed form (Section 3.5); at higher moment orders they are estimated numerically, and the variance law is exact in either case.

3.5. The projection variances in closed form

Section 3.4 rules out a weight-only quadratic form for ζ_1. What it does not rule out, and what holds, is a cubic one. The ingredient is already in the literature. Lemma 4 of [1] gives the exact second moment of two reconstructed Pauli coefficients under local shadows,

$$\mathbb{E}[\hat{x}_P\, \hat{x}_Q] = 3^{\,|\mathrm{supp}(P)\,\cap\,\mathrm{supp}(Q)|}\ \mathrm{Tr}(\rho\, P\ominus Q)\,,$$

when P and Q agree on every qubit where both are non-identity and zero otherwise, where $\hat{x}_P = \mathrm{Tr}(GP)$ and $P \ominus Q$ cancels the shared non-identity factors. We claim no novelty for this identity; what follows is its evaluation.

Since $\mathrm{Tr}(G\rho) = 2^{-n} \sum_s \langle P_s \rangle \hat{x}_s$, the first projection variance follows by contraction, and since the k-argument kernel of Section 3.1 is multilinear in the snapshots, every ζ_c is a quadratic functional of the same second-moment matrix. Two consequences precede the evaluation.

First, the diagonal $s = s'$ of that contraction is exactly the weight-only form of Section 3.4, with $c_w = 3^w/4^n$, so the residual that form leaves is exactly the off-diagonal sum. The shortfall identified there is not an open feature of the algebra.

Second, because only second moments enter, the projection variances do not distinguish local Haar single-qubit shadows from local Pauli-basis shadows: the two ensembles have entrywise identical second-moment matrices, so ζ_c agrees for every c and every k. Every statement below holds for either.

Everything in this subsection concerns $k = 2$. The higher moment orders are treated numerically as before; their projections do not reduce the same way, and we say what is known about them at the end.

a. The second projection variance is a spectral sum. The two-copy kernel is diagonal in the Pauli basis, and collecting the compatible pairs by the string they leave behind gives, for any state,

$$\zeta_2 \;=\; 7^n \sum_P \langle P\rangle^2 \left(\tfrac{1}{14}\right)^{|P|} \;-\; \mathrm{Tr}(\rho^2)^2,$$

the sum running over all 4^n Pauli strings with $|P|$ the weight. Only the identity term is state-independent, so the base of ζ_2 is never below 7, and a counting argument over compatible pairs bounds it above by 17/2. At the maximally mixed state every other term vanishes and the expression returns $7^n - 4^{-n}$, recovering Section 2.5. What sets the base is how the state's Pauli weight is distributed: a spectrum spread over all 4^n strings leaves the identity term dominant and the base at exactly 7, while a spectrum concentrated on a stabilizer group lifts it.

b. Evaluation for the noisy-pure ensemble. Averaging over the ensemble of Section 2.5, $\rho = (1-q)|\psi\rangle\langle\psi| + qI/2^n$ with $|\psi\rangle$ Haar, requires only the second and third Haar moments of Pauli expectations, $\mathbb{E}[\langle P\rangle^2] = 1/(d+1)$ and $\mathbb{E}[\langle P_u\rangle\langle P_s\rangle\langle P_{s'}\rangle] = 2/((d+1)(d+2))$ when $P_sP_{s'} = P_u$, together with four counts over compatible Pauli pairs. Writing $d = 2^n$, $u = 1-q$, and $p_2 = \mathrm{Tr}(\rho^2) = u^2 + q(2-q)/d$,

$$\zeta_1 = 4^{-n}\left[1 + \frac{u^2(10^n-1)}{d+1} + \frac{2u^2(4^n-1)}{d+1} + \frac{2u^3(16^n - 10^n - 2\cdot 4^n + 2)}{(d+1)(d+2)}\right] - p_2^2,$$

$$\zeta_2 = 4^{-n}\left[28^n + \frac{u^2(34^n - 28^n)}{d+1}\right] - p_2^2,$$

the second being the spectral sum above evaluated on this ensemble. Each qubit contributes one identity of weight 3^0 and three Paulis of weight 3^1, so $\sum_s 3^{|s|} = 10^n$; the compatible pairs give 16^n with that weighting and 34^n with its square, with diagonals 10^n and 28^n.

c. The threshold and what sets its base. Write β for the growth base of $\mathbb{E}[\mathrm{Tr}(G\rho)^2]$, which is also the base of ζ_1. Since $M^* = \zeta_2/(2\zeta_1)$, its base is $\mathrm{base}(\zeta_2)/\beta$ exactly, and both factors are spectral functionals of the state rather than universal constants. For the ensemble of Section 2.5 the closed forms give $\beta = 5/4$ and $\mathrm{base}(\zeta_2) = 7$, so

$$M^*(n) \;\longrightarrow\; \frac{1}{2(1-q)^2}\left(\frac{28}{5}\right)^n, \qquad \frac{28}{5} = 5.6,$$

with base and prefactor both exact.

The same two functionals can be evaluated for the other ensembles this paper uses, and they do not all agree:

state family	β	base(ζ_2)	M^* base
noisy-pure ($q = 0.1$)	$5/4$	7	$28/5 = 5.60$
Haar-pure	$5/4$	7	$28/5 = 5.60$
low-rank (rank 2)	≈ 1.21	7	≈ 5.8
product, GHZ	$3/2$	$15/2$	5.00

The last row is exact and instructive. For a pure product state the spectral sum is $\sum_P \langle P\rangle^2 14^{-|P|} = (15/14)^n$, giving $\zeta_2 = (15/2)^n - 1$, and for GHZ the stabilizer group splits into even-weight Z strings and weight-n strings, giving $\zeta_2 = \frac{1}{2}\left[(15/2)^n + (13/2)^n\right] - \frac{1}{2}$. Purity alone does not determine the base: a Haar-random pure state gives 7 and a pure product state gives 15/2. What distinguishes them is the weight profile of the Pauli spectrum, which the kernel $14^{-|P|}$ reads off. The value 28/5 is therefore the spread-spectrum case, not a universal constant, and structured stabilizer states sit on a different branch at 5.

d. Higher moment orders. The multilinearity argument above shows every ζ_c depends only on the second-moment matrix, so closed forms exist in principle at $k \geq 3$ as well. We did not obtain them: at $k = 3$ and 4 the state-dependent contributions dominate the state-independent ones, so the kernel bases are not read off the maximally mixed values, and the required counts over compatible tuples are beyond what we derived. The $k \geq 3$ coefficients used below are estimated numerically.

The fitted base drifts with the fitting window, from 5.14 over $n = 2$ to 7 to 5.60 over $n = 4$ to 9. The closed forms reproduce that drift exactly, returning 5.18, 5.36 and 5.60 over the same windows with no sampling anywhere, so the drift is a finite-size property of the fit and 28/5 is what it drifts toward.

3.6. The α transition

The projection variances for noisy-pure states, estimated over this range, give clean exponential scalings — sampled-fit constants, retained here for the contrast drawn in Section 3.5, which quotes 5.14 for the same window from the regression on $M^*(n)$ and 5.18 from the closed forms. Over $n = 2$ to 7,

$$\zeta_1 \approx 0.63 \cdot (1.35)^n, \qquad \zeta_2 \approx 1.10 \cdot (6.93)^n, \qquad M^* \approx 0.87 \cdot (5.15)^n,$$

and over the wider range $n = 2$ to 9, $M^* \approx 0.76 \cdot (5.345)^n$. The base of ζ_1 is mildly curved, reading approximately 1.35 at small n and 1.30 over the full range, which accounts for the difference between the two fits.

The kernel variance grows roughly five times faster per qubit than the linear variance. Their ratio, the threshold M^*, therefore grows approximately exponentially in n over the range we fit ($n = 2$ to 9). This is the mechanism, and within that range its consequence is direct: at a fixed budget, increasing n pushes M below M^*, the higher-order term takes over, and the effective exponent migrates from 1/2 toward 1, and past it for higher k.

Defining α as the effective exponent — the negative slope of $\log \mathrm{RMSE}$ against $\log M$ fitted by least squares over the budget range actually used, as in Appendix C, so it is a window-averaged rather than a pointwise quantity and the window itself shifts with n:

n	$M^*(n)$	predicted α	measured α
2	≈ 25.6	0.501	0.495 ± 0.013
4	≈ 549	0.528	0.528 ± 0.012
9	$\approx 3.0 \times 10^6$	0.998	1.006 ± 0.023

The $M^*(n)$ column is the exact threshold M^* evaluated at each n from the closed-form projection variances, not the fitted law above evaluated at n. The predictions are computed from the projection variances — closed-form at second order, estimated from shadow snapshots at higher orders — fed into the exact variance formula. Nothing is fitted to the α data (Fig. 1). Across every system size and moment order for which we have measurements, the derived law matches: 8 of 8 within two standard errors at $k = 2$, 6 of 7 at $k = 3$, where the single miss sits on the two-sigma boundary and is sensitive to the random seed, and 5 of 5 at $k = 4$.

The exact combinatorial structure is necessary, not a convenience, although $k = 2$ is the wrong place to look for the evidence. There the large-M truncation of Section 3.2, $\mathrm{Var} \approx 4\zeta_1/M + 2\zeta_2/M^2$, tracks the exact variance to within 0.05% across the budgets used here and reproduces the measured α at 8 of 8, indistinguishable from the exact law: at $k = 2$ the second projection and the full kernel variance are the same quantity, so the truncation discards nothing. They separate at higher orders. Keeping the same two terms with their correct coefficients, $k^2\zeta_1/M + \frac{1}{2}k^2(k-1)^2\zeta_2/M^2$, gives 15 of 20 across $k = 2, 3, 4$ against the exact law's 19 of 20, and misses $k = 3$, $n = 8$ by twenty-six standard errors — by then ζ_3/ζ_2 exceeds 10^4 and the discarded terms carry the variance.

A log-linear regression of $M^*(n)$ over $n = 2$ to 9 returns a base of 5.34 with a 95% confidence interval of [5.14, 5.54] and a prefactor of 0.75 ± 0.09 (regression standard errors, treating each $M^*(n)$ as fixed); ζ_1 and ζ_2 fit to bases 1.30 [1.26, 1.34] and 6.94 [6.90, 6.98]. The ζ_2 base sits just below the exact per-qubit second moment 7 at the maximally mixed state (Section 3.3) and barely moves across the fitting windows; the base drift is carried by ζ_1. The residuals carry mild upward curvature, so the fitted base drifts with the window in the manner Section 3.5 anticipates; Appendix C gives the residual, window, and leave-one-out detail.

A fixed-exponent bound RMSE $\leq C_n/\sqrt{M}$ is not wrong; it is a bound and it holds, and the tighter two-term treatments do not assume a fixed exponent either [19]. What a fixed-exponent summary describes is the wrong regime for the systems being run. At nine qubits and the budgets accessible on current hardware the estimator sits near $\alpha = 1$, where extra shots buy error reduction faster than $1/\sqrt{M}$ but from a variance already exponentially large, so any extrapolation presuming $\alpha = 1/2$ errs in a predictable direction. At fixed n the linear term dominates once the budget grows without bound, so the asymptotic scaling remains the familiar square-root law: the migration reported here is a finite-budget effect over the window hardware actually reaches, not a change in asymptotic sample complexity.

Having characterized the single-copy cost exactly, we turn to the collective route, where the error has a different character entirely.

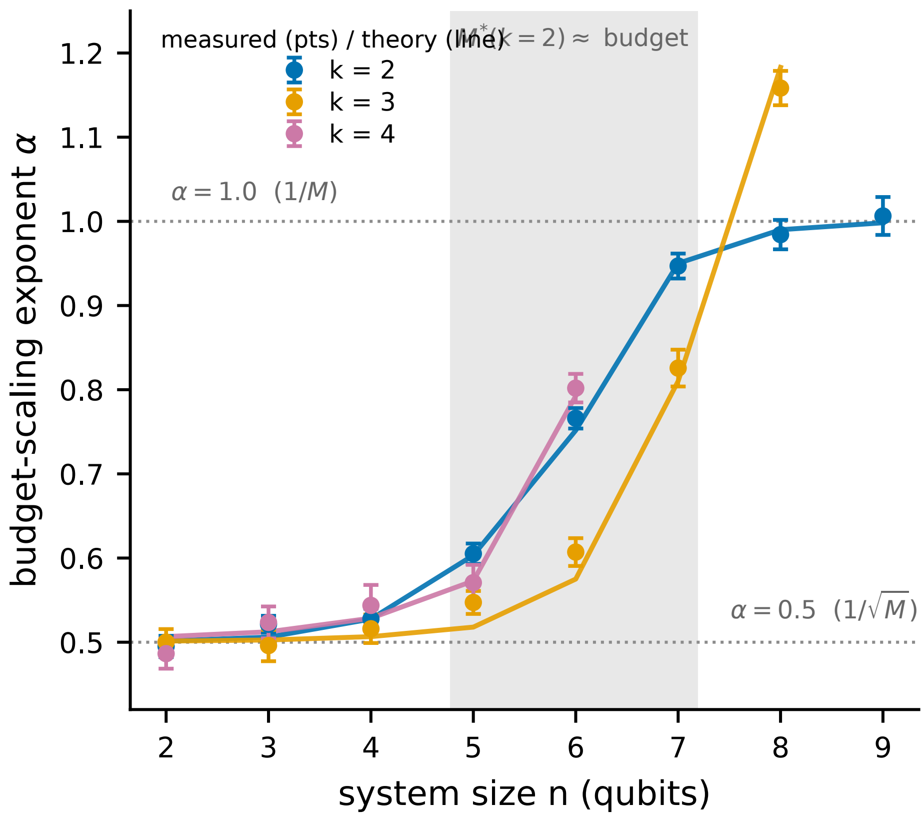

FIG. 1. The effective budget-scaling exponent transition, the paper's central phenomenon. Measured budget-scaling exponent α (RMSE $\propto M^{-\alpha}$, fitted over budgets 2000–128000; markers with bootstrap standard errors) versus system size n for moment orders $k = 2, 3, 4$, with the derived-law prediction (lines). *The prediction curves have no parameters fitted to the plotted points, computed from projection variances fed into the exact variance formula of Sec. 3.1, and are not fits to the plotted points.* As n grows at fixed budget and M falls below the threshold M^*, α migrates continuously from 1/2 toward 1, and past it for the higher moment orders.

4. THE COLLECTIVE ROUTE: TWO EXACT BIAS LAWS

At nonzero noise, once the shot budget is large enough that the finite-shot term has decayed, the collective route's error is dominated by a bias rather than a variance; at zero noise or small budgets the finite-shot variance dominates instead. The bias is independent of the shot budget, a prediction Section 5.3 tests directly and which holds. Its functional form is fixed by the geometry of the noise channel, its magnitude by geometry and rate together, and two channel classes give two exact laws.

4.1. Global depolarizing noise: linear in g, with no compounding

Let the kn-qubit collective register be subject to global depolarizing noise at rate g, so that $\rho^{\otimes k} \mapsto (1-g)\rho^{\otimes k} + g\,\mathbb{I}/d^k$ with $d = 2^n$. Then

$$\mathrm{Tr}\Big(C_k\left[(1-g)\rho^{\otimes k} + g\,\tfrac{\mathbb{I}}{d^k}\right]\Big) = (1-g)\,\mathrm{Tr}(\rho^k) + g\,\frac{\mathrm{Tr}(C_k)}{d^k} = (1-g)\,\mathrm{Tr}(\rho^k) + g\,2^{\,n(1-k)},$$

using $\mathrm{Tr}(C_k) = d$ from Section 2.3. Hence

$$\boxed{\mathrm{bias} = g\,\big|\mathrm{Tr}(\rho^k) - 2^{\,n(1-k)}\big|}$$

At a fixed g the bias is linear in g and does not compound across qubits: the single-cycle property of C_k collapses the compounding that a per-qubit form $1-(1-g)^{kn}$ would predict, and which overestimates the bias by a factor of five to fourteen over the range swept here. The two forms are parametrically different in n, the compounding form saturating toward unity while the true bias stays linear. This holds at fixed g; on hardware the effective global rate may itself grow with register size, depth, or gate count.

4.2. Per-qubit channels: the noise relabels the state

Let $\mathcal{E}$ be any single-qubit channel borne by the collective route and not by the single-copy route, applied to every qubit of every copy held for the test, with ρ the reference state and $\mathrm{Tr}(\rho^k)$ the estimand. That $\mathcal{E}$ is specific to this route is a modeling choice rather than a derivation, and it is what makes the quantity below a bias.

The identity needs one further precondition, and it is narrow: the k copies must be independent and must undergo $\mathcal{E}^{\otimes n}$ *before* an otherwise ideal cyclic-permutation measurement. It does not cover noise inside the cyclic test — imperfect controlled-permutation gates, ancilla decoherence, readout error — nor correlated noise acting across copies, nor any channel acting during or after the circuit. Those perturb the measured observable or the instrument rather than the state being measured, and the relabeling does not apply to them. Within that precondition,

$$\mathcal{E}^{\otimes nk}\big(\rho^{\otimes k}\big) = \big(\mathcal{E}^{\otimes n}(\rho)\big)^{\otimes k} = \sigma^{\otimes k}, \qquad \sigma \equiv \mathcal{E}^{\otimes n}(\rho),$$

and the cyclic test of a product of identical states returns the k-th moment of that state:

$$\boxed{\text{measured} = \mathrm{Tr}(\sigma^k), \qquad \sigma = \mathcal{E}^{\otimes n}(\rho)}$$

so the bias is $\left|\mathrm{Tr}(\sigma^k) - \mathrm{Tr}(\rho^k)\right|$. The noise does not corrupt the measurement in the usual sense; it relabels which state is measured, and that substitution is the entire bias. A channel common to both routes would bias the single-copy route by the same substitution and cancel from the comparison, carrying no information about which route has the lower RMSE at equal state-copy budget; the law requires that $\mathcal{E}$ be borne by the collective route alone. Preparation noise is not excluded but already absorbed into ρ by the ensemble of Section 2.5, which both routes see and which contributes no bias.

4.3. Verification

Both laws are exact identities rather than approximations, and we verified them accordingly: against explicit construction of C_k and the noise-damaged k-copy state, at $n = 2, 3$ and $k = 2, 3, 4$, for global depolarizing, amplitude damping, and dephasing, agreeing to approximately 10^{-15}. We take the single-qubit Kraus operators at rate g in the standard form: amplitude damping as $K_0 = \mathrm{diag}(1, \sqrt{1-g})$ and $K_1 = \sqrt{g}\,|0\rangle\langle 1|$; dephasing as $K_0 = \sqrt{1 - g/2}\,\mathbb{I}$ and $K_1 = \sqrt{g/2}\,Z$; and global depolarizing in the closed form of Section 4.1. Each is applied to every qubit of every copy. They continue to hold at three times the noise rates in which they were derived, and on state ensembles they had never been tested against: Haar-pure ($q = 0$); low-rank, $\rho = GG^\dagger$ with G a $2^n \times 2$ complex Ginibre matrix normalized to unit trace, whose spectrum is not the rank-one-plus-identity form the noisy-pure closed forms assume; and GHZ-noisy, $(1-q)|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}| + q\,\mathbb{I}/2^n$ at $q = 0.15$ with $|\mathrm{GHZ}\rangle = (|0\cdots 0\rangle + |1\cdots 1\rangle)/\sqrt{2}$. An exact identity is not expected to degrade under extrapolation, and it does not.

With the single-copy variance and the collective bias both known exactly, the crossover follows by setting one against the other.

5. THE CROSSOVER

5.1. The law

The single-copy error grows exponentially in n and falls with the budget. The collective error is a bias floor plus a finite-shot variance: the floor is bounded above by $1 - d^{1-k}$ with $d = 2^n$, hence by unity, and is independent of the budget, while the shot term vanishes as the budget grows. The crossover size n^* is where the single-copy error rises above the collective error,

$$\underbrace{\sqrt{\mathrm{Var}(U_M)}}_{\text{exponential in } n,\ \text{falls with } M} \quad = \quad \underbrace{\sqrt{\mathrm{bias}(g,k,n)^2 + \frac{1-\mu^2}{N}}}_{\text{bounded floor + shot noise}}$$

Here μ is the noisy collective signal the cyclic test returns and $N = \lfloor M/k \rfloor$ is the number of k-copy measurements a copy budget of M allows. The shot term is the variance of the bounded cyclic-test outcome and falls away as the budget grows, leaving the bias floor as the large-budget limit. Every quantity on both sides comes from Sections 3 and 4; nothing is fitted to the data being predicted, and the laws take the state's projection variances and the noise channel as inputs.

The bounded-floor claim is a statement about bias in *estimation* at fixed additive precision ε: the collective-test outcome is a bounded random variable, so estimating $\mathrm{Tr}(\rho^k)$ to fixed ε costs $O(1/\varepsilon^2)$ shots at every n, and the collective error saturates at the floor rather than growing. It is not a claim that collective measurement is cheap for all tasks: for a task whose target precision must itself shrink exponentially in n, [6] proves that exponentially many samples are required even with two copies. Section 7 gives the quantitative reconciliation.

5.2. Validation on 83 cells and four ensembles

Across 83 cells spanning moment order $k \in \{2, 3, 4\}$, three noise models (depolarizing, amplitude damping, dephasing), noise rates from 0 to 0.3, sizes $n = 2$ to 9, four budget multipliers, and four state ensembles, the criterion predicts 82 of the 83 simulated crossovers within one qubit and 73 of 83 exactly (Figs. 2 and 3).

The comparison is asymmetric between the routes in two respects, both deliberate. The first is evidential: the single-copy side of each cell is a forward simulation of the estimator, fresh snapshots reduced by the U-statistic, so the variance law is tested against data it did not generate, while the collective side is the exact bias law evaluated analytically plus its shot noise.

The second is the noise model, and it is the stronger assumption. The channel of Section 4 degrades the collective route only; the single-copy route is the ideal, noiseless statistical baseline, on the reading that a joint k-copy measurement is the harder thing to implement cleanly. Real shadow readout is not noiseless, and Section 6.3 finds readout to be the dominant error on the device we tested, so this understates the single-copy cost — in the direction that costs us, and we have measured by how much. Simulating readout on both routes at the per-qubit rates measured in Section 6.3, and correcting the collective parity for readout exactly through its outcome-dependent per-pair contraction rather than a blanket factor, moves the crossover to smaller n in every resolved cell: by one to two qubits when both routes calibrate readout out, and further when both leave it uncorrected. The sizes reported here are therefore a conservative upper bound under this noise model. Under uncorrected readout the single-copy U-statistic acquires a bias that does not shrink with the copy budget, unlike its noiseless variance, so both routes then carry error floors and the comparison changes character. What the asymmetry does shape is the dependence of n^* on the noise rate, which is a property of the collective side by construction.

These cells demonstrate consistency with the derived identities; the out-of-ensemble transfer is the evidence that the law generalizes. What transfers is the structure of the formula, since the projection variances are re-estimated for each ensemble, so this is not zero-shot prediction from universal constants. Three of the four ensembles — Haar-pure, low-rank, and GHZ-noisy — were never used in deriving the law. GHZ states are maximally non-Haar-typical, and we included them because an ensemble-tuned theory ought to fail on them; the theory's accuracy there is within a factor of about 1.7 of its accuracy elsewhere, the same order of magnitude rather than a breakdown.

5.3. Three qualitative predictions, and the one we got wrong

Two of the three we predicted correctly in advance; the third we got backwards, and the law corrected us.

Higher noise moves the crossover later on the channels and rates we swept: a larger bias

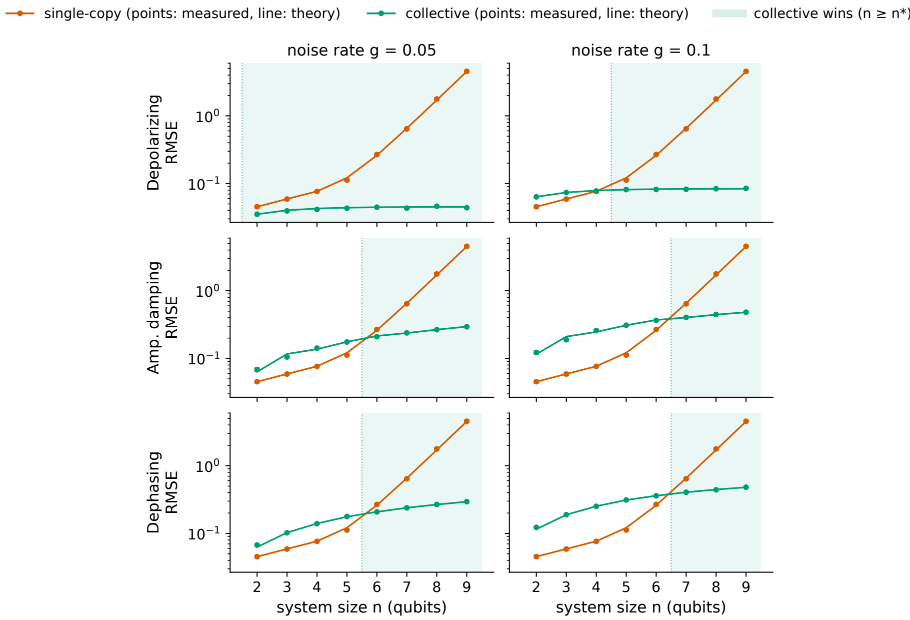

FIG. 2. The crossover for purity ($k = 2$). Single-copy classical-shadow RMSE (vermillion) and collective two-copy RMSE (green) versus system size n at a fixed copy budget, for three noise channels. *The theory curves are predictions with no parameters fitted to these points, from the exact variance and bias laws (Secs. 3–4), not fits to these points.* The crossover n^* is where the exponentially growing single-copy error overtakes the collective error, a bounded bias floor plus a finite-shot variance that vanishes as the budget grows.

floor takes longer for the exponentially growing single-copy error to climb over. A larger budget moves it later too, because the single-copy error falls with budget while the bias floor does not; the collective RMSE plateaus exactly at the predicted floor as the budget grows while the single-copy RMSE keeps falling, a direct test of the bias-versus-variance distinction on which the whole crossover rests.

Higher k also moves the crossover later on the families we tested, and this is counterintuitive: a naive variance argument says higher moments compound the single-copy variance faster and should cross earlier. The law explains the reversal. $\mathrm{Tr}(\rho^k)$ shrinks with k, so the absolute bias floor the single-copy error must climb over is lower, and single-copy holds out to larger n. We report this as an observation about the ensembles and channels swept here rather than a general rule: $|\mathrm{Tr}(\sigma^k) - \mathrm{Tr}(\rho^k)|$ need not fall monotonically with k for an arbitrary spectrum and channel. No universal effective attenuation rate fits the measured biases either, because the bias is not a rate: it is however much the channel deforms that

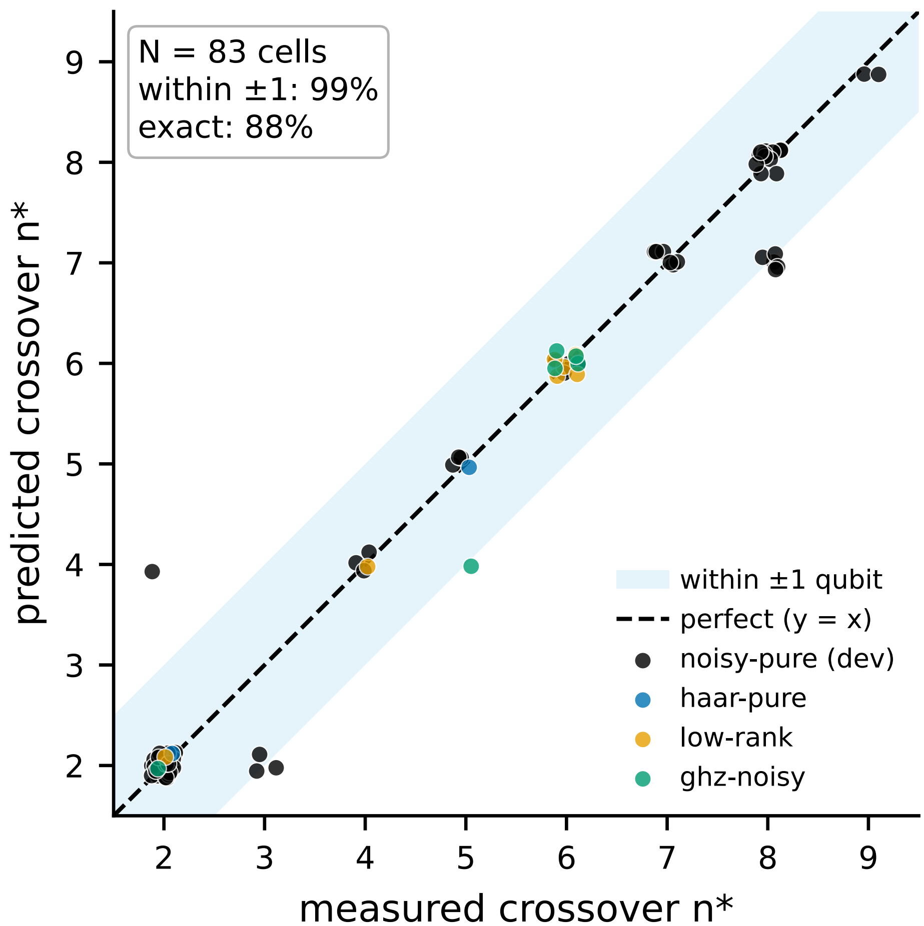

FIG. 3. Crossover validation across 83 cells. Predicted versus Monte-Carlo-observed crossover size n^* over moment orders $k \in \{2, 3, 4\}$, three noise models, and four state ensembles (three unseen by the theory). Accuracy is reported in Sec. 5.2. The cells are simulation, not hardware; for the higher moment orders the collective side is an idealized bounded-outcome primitive carrying no gate, depth, or routing cost.

particular state's spectrum, and two states of identical purity can be deformed by different amounts. Purity ($k = 2$) crosses at $n \approx 6$ to 7 under realistic noise, while $k = 3$ and $k = 4$ hold out to $n \approx 8$ — sizes for the idealized bounded-outcome test of Section 2.3, whose realization costs about an order of magnitude more two-qubit gates than the $k = 2$ test, plus an ancilla and routing. The signal-against-a-fixed-floor argument dominates the variance-compounding argument.

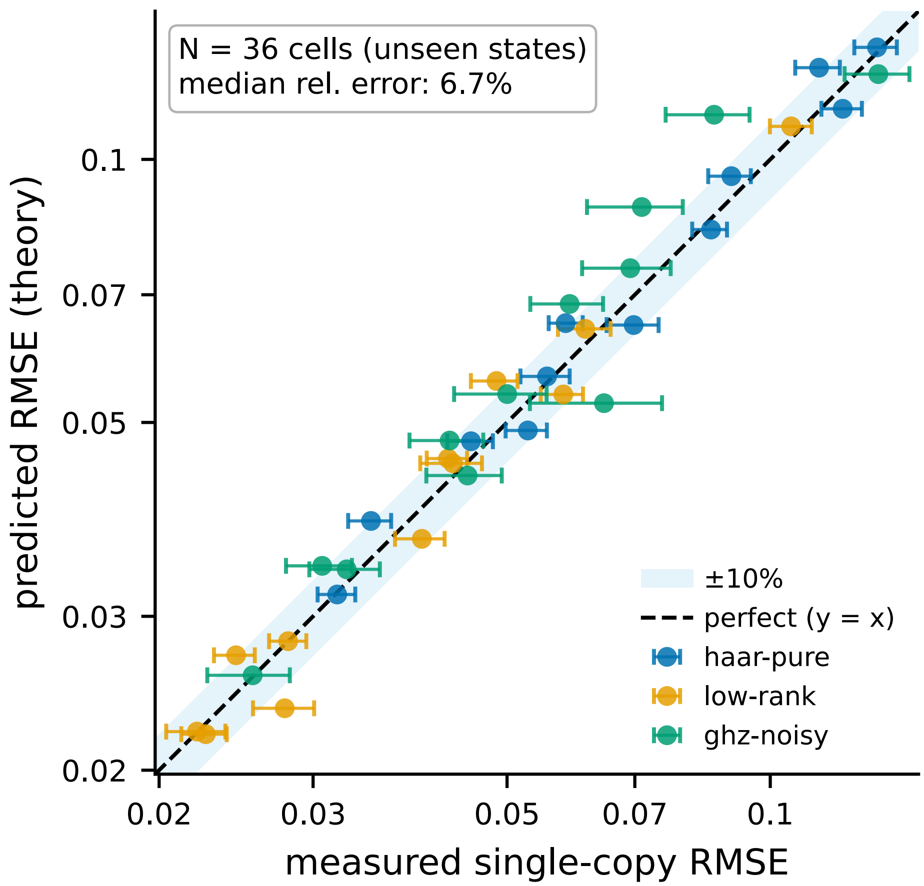

FIG. 4. Out-of-ensemble validation of the variance law. Predicted versus measured single-copy RMSE on three ensembles the theory never saw (Haar-pure, low-rank, GHZ-noisy). The scatter about the diagonal is finite-trial estimation noise (median relative deviation 6.7%; Sec. 8), not a systematic error.

5.4. The exponential wall

The practical statement is a single sequence. Using the exact copy-fair estimator on noisy-pure states at a fixed copy budget, the single-copy purity RMSE runs

$$0.043 \;\rightarrow\; 0.072 \;\rightarrow\; 0.270 \;\rightarrow\; 1.62 \;\rightarrow\; 11.98$$

at $n = 2, 4, 6, 8, 10$, growing by a factor of about two per qubit on average and accelerating to roughly 2.7 per qubit over the last step, against a true purity of approximately 0.81 (Fig. 5).

At ten qubits the single-copy estimate carries an error fifteen times larger than the quantity it is estimating. The collective route stays bounded over the same range, because its error is a bias floor and the floor cannot exceed $1 - d^{1-k} < 1$: both $\mathrm{Tr}(\sigma^k)$ and $\mathrm{Tr}(\rho^k)$ lie in $[d^{1-k}, 1]$, so their difference is bounded by the width of that interval. What is universal is the weaker statement that the floor stays below unity, and that is what the crossover argument uses.

This speaks to the resource-cost concern behind the choice in [5] to keep its measure-first protocol single-copy. The entangling overhead that motivates such caution is real, and it is not the binding constraint. The binding constraint is that single-copy estimation of nonlinear functionals by the local-randomized-measurement shadow U-statistic we study does not survive to the sizes they are entering — a statement about this protocol, not about every single-copy strategy. Whether the collective route survives on a real device is the question we take up next.

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6. HARDWARE: A COMMITTED-IN-ADVANCE TEST ON RIGETTI CEPHEUS-1-108Q

6.1. Platform, and a disclosure about the execution path

All measurements were submitted to the Rigetti Cepheus-1-108Q backend as provisioned by the platform, a 108-qubit superconducting processor built from twelve interconnected nine-qubit chiplets [10]. Published median fidelities at the time of the experiments were 99.1% for the two-qubit CZ gate and 99.9% for single-qubit gates. Readout error is absent from that release, a fact that matters in Section 6.3. These figures are the specification as published at that date; per-job backend and calibration metadata were not available through the execution path we used.

Access was obtained through the Open Quantum platform operated by Quantum Rings [9], on the Public Tier. That tier does not route jobs directly to the vendor: circuits are routed to distributed operators on the Quantum Compute subnet (SN48) of the Bittensor network, operated by qBitTensor Labs, with validators performing spot checks that circuits ran on the appropriate target QPU [9]. We therefore cannot independently verify the

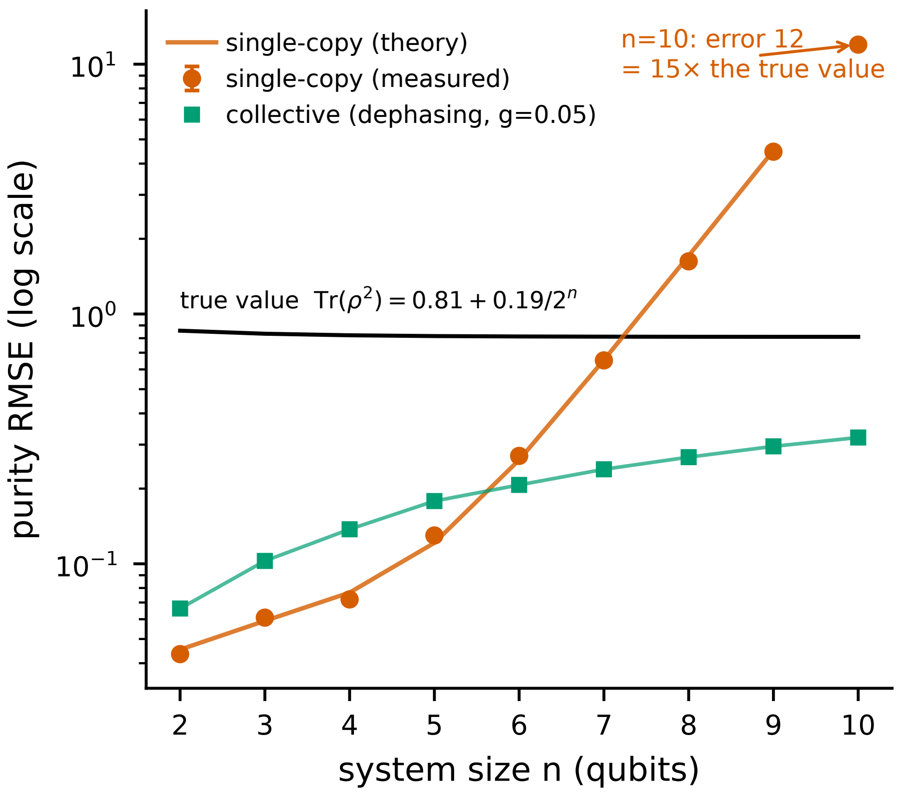

FIG. 5. The exponential wall. Single-copy purity RMSE versus n out to $n = 10$ at a fixed copy budget of $M = 2000$ (log scale; measured points with 68% CIs, theory line with no parameters fitted to the crossover data; noisy-pure ensemble, collective route under dephasing at $g = 0.05$), with the exact per-n true purity $\mathrm{Tr}(\rho^2) = 0.81+0.19/2^n$ marked (0.858 at $n = 2$, approaching 0.81). By $n = 10$ the single-copy error reaches ~ 12, about fifteen times the quantity being estimated, while the collective error (green) stays bounded.

execution path of any individual job. This does not affect the characterizations in Sections 6.2 and 6.3, which are internally consistent, but it is a candidate contributor to the session-to-session variability of Section 6.4 that we cannot rule out, and we list it as such in Section 8. Jobs were submitted as OpenQASM 3 with physical-qubit addressing, which the platform required. Registers, protocol, license conditions, and the credit ledger are in the supplementary material.

6.2. The modeled entangling overhead is small for the tested circuits

The destructive SWAP test at $n = 2$ transpiles onto physical qubits $\{0, 1, 9, 10\}$ using four CZ gates and zero routing SWAPs; the GHZ ladders at $n = 3$ and $n = 4$ likewise map with zero routing at $3n - 2$ CZ gates, and the ladders nest. CZ error on these qubits was not measured directly: the platform compiler resynthesized our characterization echo and optimized its gates away, and the verbatim box that would have prevented this failed at execution. We therefore assume CZ error at the published median of 0.9%, an average gate infidelity r corresponding under the depolarizing convention $\Lambda(\rho) = (1 - p)\rho + p\,\mathbb{I}/d$ to a channel parameter $p = \frac{d}{d-1}r = \frac{4}{3}r = 0.012$ for a two-qubit gate ($d = 4$).

On that assumption gate error accounts for roughly 0.042 of the purity deficit at $n = 2$ against a total deficit of 0.29, about a seventh (roughly 14%). At the lower median the vendor has since published for this backend (98.71%, $r = 0.0129$, $p = 0.0172$) it rises to about 0.057, close to a fifth of the measured deficit (0.20). This is a device-wide median rather than an edge-specific measurement, and a depolarizing model captures neither coherent error nor leakage, so the figure is a scale rather than a budget; under either median it is too small to account for the deficit.

The resource-demand concern cited in [5] therefore does not materialize for the destructive SWAP test on a square-lattice device. That is a property of structured states on a matched topology rather than of the test in general: generic states at $n = 4$ require 46 CZ gates on this topology, 26 intrinsic plus 20 routing-attributable, against the GHZ ladder's 10 with zero routing, and we restrict the hardware series to GHZ ladders accordingly. This is a $k = 2$ statement; the ancilla-based Hadamard test required for $k \geq 3$ (Section 2.3) carries a cost we have not measured.

6.3. Consistent with readout and SPAM error dominating, and not in the datasheet

Our locked prediction, computed from the published gate fidelities and an assumed 2% readout error and committed to version control before the job was submitted, was a Bell-state purity of 0.8932. The device returned

$$0.7184 \quad [0.6992,\, 0.7376] \ \ \text{(95\% bootstrap CI, 5000 shots)},$$

against a true purity of exactly 1, overshooting by more than seventeen standard errors — conditional shot noise, hence a lower bound on the total uncertainty (Appendix C). The estimand here is the target the circuit was written to prepare, so every deviation from it is charged to the global-depolarizing channel of Section 4.1, which has no single-copy reading to be relabelled into.

The output structure is consistent with the intended circuit: the four dominant outcomes are exactly the ideal support of the Bell SWAP test, with noise leaking approximately 25% of the weight into the remaining twelve. Inverting the bias law of Section 4.1 gives an effective global depolarizing rate $g = 0.375$, some two and a half times the $g = 0.142$ the locked prediction implied.

A single measurement cannot separate gate error from readout error, so we characterized readout separately, using computational basis states prepared with X gates only and no two-qubit gates. What this isolates is combined state-preparation-and-measurement error rather than assignment error alone, so "readout" here should be read as SPAM:

qubit	$P(1 \mid 0)$	$P(0 \mid 1)$
q_0	9.3%	8.9%
q_1	0.7%	7.8%
q_9	6.7%	9.8%
q_{10}	3.2%	6.0%

We write q_N for the physical qubit addressed as `$N` in the platform's OpenQASM 3 syntax. The mean is approximately 6.5% per qubit, roughly three times the assumed value, and asymmetric in the direction expected from T_1 decay during readout. Feeding it back into the model moves the predicted purity from 0.89 to approximately 0.75, against the measured 0.7184. Readout on this device is also correlated: q_0's $P(1|0)$ rises from 1.6% with idle neighbors to 16.9% with excited ones, and including that crosstalk closes the residual, predicting 0.7163 against the measured 0.7184. The correlation saturates with excitation weight rather than accumulating, so it must be measured at each weight rather than extrapolated; the weight-resolved measurements are in the supplementary material. The input that was wrong is precisely the one the specification does not publish.

6.4. Cross-session variation is the leading explanation for the gap

With readout on the GHZ ladder measured in the same session rather than assumed, we ran a bracketed experiment (Fig. 6): opening calibration, predictions locked and committed to version control from it, SWAP measurements, closing calibration. Within-session drift measured approximately 1% per qubit, so the device was stable while the experiment ran. All three cells nonetheless fell below both prediction bands rather than between them, and the degradation was non-monotonic: $n = 3$ came out worse than $n = 4$, despite the $n = 4$ register containing the $n = 3$ one. Neither readout, which scales with the $2n$ measured qubits, nor CZ count, which grows with n, can produce that ordering. The band-by-band comparison is in the supplementary material.

The cross-session picture explains it. Running byte-identical circuits on the same physical qubits across three sessions, the $n = 3$ register healed ($0.317 \to 0.378 \to 0.587$) while $n = 4$ drifted down ($0.431 \to 0.420 \to 0.382$), and the ordering between them flipped. Within-session drift propagates through the device model to about 0.01 in purity; cross-session drift is approximately 0.2, roughly twenty times larger on that like-for-like comparison. Identifying the non-monotonicity as a drift artifact required running the same circuit three times over several days, which is not standard practice.

The single-copy route failed the same way. A fifteen-basis $n = 2$ anchor, whose U-statistic our shadow pipeline predicted at 1.500 ± 0.028 from the same device parameters, returned 1.344 with a 95% bootstrap interval of [1.266, 1.433] over 9000 snapshots: 5.6 times the spread the model predicts, with disjoint intervals, and no simulated run of the locked model reaches the measured value. That statistic is not a purity, and the anchor tests this pipeline rather than the standard i.i.d. estimator, for reasons given in the supplementary material. We do not adjust the model to fit. Three signals therefore fail together in one session, and that two protocols with different circuits and different estimators both miss their locked predictions is what makes a session-dependent cause, rather than any protocol-specific defect, the parsimonious explanation.

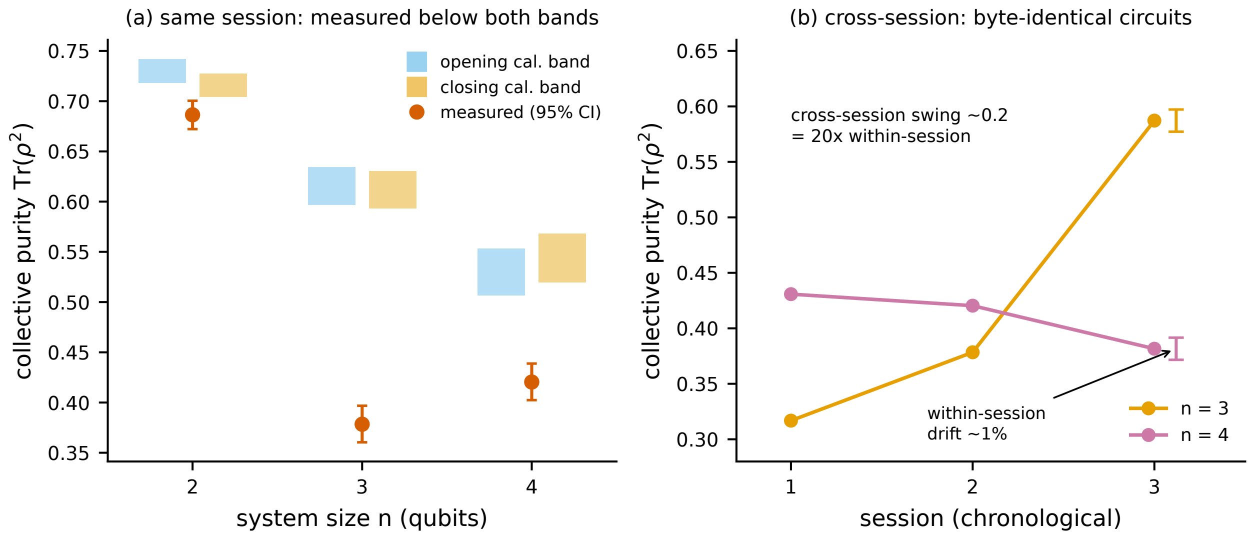

FIG. 6. Hardware: cross-session variation is the leading explanation for the gap. (a) Measured collective purity (vermillion, 95% CI) at $n = 2, 3, 4$ against the same-session prediction bands committed in advance from the opening (blue) and closing (orange) readout calibrations; every measurement falls below both bands. (b) The same byte-identical circuits across three sessions: the $n = 3$ register healed ($0.32 \to 0.59$) while $n = 4$ drifted down ($0.43 \to 0.38$), and the ordering flipped. Cross-session drift (~ 0.2) is about $20\times$ the within-session drift (scale bar).

6.5. The elimination ledger

Each candidate mechanism was addressed with a quantitative argument rather than by absence of evidence: four are excluded outright, and three are bounded well below the observed deficit.

mechanism	how it was bounded or excluded
Within-session drift	Bracketed calibration measured approximately 1% per qubit, against a 0.2 discrepancy.
Single-copy readout error	Directly characterized. Feeding measured values into the model does not close the gap at $n \geq 3$.
Gate count	Cannot produce $n = 3$ worse than $n = 4$, since CZ count is monotone in n.

mechanism	how it was bounded or excluded
Cross-copy readout crosstalk	Ruled out by a physical bound. Joint readout on the full $2n$ register is factorizable (TVD 0.009 at $n = 3$, 0.011 at $n = 4$), and parity-pair correlations are negligible and no stronger than non-pair correlations. Holding the measured single-qubit flip rates fixed and pushing every copy-pair to its maximum physically realizable correlation, which is bounded because $P(\text{both flip}) \leq$ min of the two marginals and the measured marginals are small, the entire achievable range of predicted purity is $[0.611, 0.669]$ at $n = 3$. The measurement is 0.378. Even maximal correlation leaves 0.23 of the deficit unexplained by this mechanism.
Coherent gate error	Randomized compiling. Pauli-twirled SWAP circuits, with a positive control confirming that the inserted gates physically execute (5.4:1 mass on the disjoint support), moved the purity by -0.013 against the 0.197 same-session $n = 4$ gap, and twirl-to-twirl scatter was shot-noise limited with zero physical scatter. A masking bound places any hidden coherent contribution at no more than 14% of the deficit. The residual is incoherent.

mechanism	how it was bounded or excluded
A specific bad qubit pair ($\{3, 12\}$)	Localization test. An $n = 3$ ladder that includes the suspect pair measured 0.581, matching the standard $n = 3$ ladder's 0.581 to well within the ≈ 0.012 shot-noise standard deviation at 5000 shots, from genuinely distinct raw data. Adding the suspect qubits changes nothing.
A fixed register-geometry fault	The $n = 4$ versus $n = 3$ ordering flips sign across sessions on identical registers. The deficit is session-dependent, not geometric.

6.6. What remains, and what we conclude

What is left is a large, incoherent, session-dependent fault that appears and disappears on the same physical qubits running byte-identical circuits. We do not identify it. Localizing it would require per-edge interleaved randomized benchmarking across many sessions, which is a research program rather than a section.

The conclusion the data support is narrower than "we found a new error mechanism" and more useful than "the device is noisy." Static device characterization did not predict collective-measurement performance over the execution path available to us and across the sessions we ran. We cannot isolate the mechanism: byte-identical circuits routed through a decentralized network may have executed on different operators' hardware, on the same hardware in different calibration states, or both. The operational consequence is the same either way. A protocol whose entire advantage rests on a bounded, characterizable bias floor cannot be deployed against a floor that moves by 0.2 between sessions, whatever moves it, and per-session verification rather than published median device parameters is what such a protocol requires.

Temporal and spatial instability of superconducting processors is documented over 22 months of observations [11, 12] and in calibration-drift analyses [13], and we claim no priority over it. What we add is a demonstration, with predictions committed in advance and a seven-mechanism ledger, that a session-dependent fault of this kind is the binding constraint specifically for collective-measurement protocols, at magnitudes that dwarf every error source such protocols are conventionally modeled with.

One further constraint affects reproducibility rather than physics. The single-copy route requires many independent random measurement bases, and on a per-circuit-priced platform without batching each is a separately billable job — one to two orders of magnitude more than the collective route used at the same shot count. That is why our hardware single-copy baseline at $n \geq 3$ is computed from independently measured device parameters rather than measured directly, with the fifteen-basis $n = 2$ anchor as the one point we could afford. That anchor did not reproduce its prediction, so the computed points are model output the hardware did not corroborate, and we present them as such. The credit accounting behind that comparison is in the supplementary material.

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7. RELATED WORK

Classical shadows and variance bounds. The classical-shadow framework [1] gives sample-complexity bounds via the shadow norm; for second-order nonlinear functionals the variance splits into a kernel term bounded by $4^{|AB|}$ and linear terms bounded by $2^{|AB|}$ [7], which states and applies the bound citing its own references, so we attribute the statement rather than its origin. These are worst-case, state-independent bounds. For local randomized measurements, Huang, Wang and Xu [22] give worst-case bounds whose exponential base depends on the measurement ensemble, for the related task of distributed inner-product estimation; these remain envelopes rather than evaluated coefficients or a threshold. The exact finite-M decomposition is the standard Hoeffding identity [17, 18], already stated as an equality for shadows by [1] and again by Elben et al. [19], who read the two-regime scaling off it. Reference [19] bounds ζ_1 and ζ_2 rather than evaluating them; [1] does supply the exact second moment, in its Lemma 4, but applies it to worst-case sample complexity rather than evaluating the coefficients for a state or an ensemble. Neither gives a threshold in M, and separate upper bounds constrain the ratio not at all, which is why the threshold does not follow from them. What this work adds is the evaluation of Section 3.5, the resulting threshold, the spectral functionals fixing its growth base, and the comparison against the collective route. Alternative shadow ensembles [8] target different observable classes.

U-statistics and the Hoeffding decomposition for quantum moments. Straeter, Tsesmelis and Kwek [14] construct unbiased U-statistic estimators for the partial-transpose moments p_2 and p_3 from randomized homodyne data and derive their variance via a Hoeffding decomposition, obtaining sample-complexity bounds for a p_3-PPT entanglement criterion. This is the closest methodological neighbor to Section 3; the differences are the platform, the target (entanglement detection rather than moment estimation and the collective-measurement trade-off), and the result, since their projection variances are bounded rather than evaluated for a state and no change of scaling regime is considered. We claim no novelty for the Hoeffding technique itself.

Crossovers and hardware comparison. A crossover in shot cost between Pauli and Clifford shadow ensembles has been reported for multipartite entanglement witnesses [15]; that is a crossover in the choice of measurement ensemble, not between single-copy and collective measurement. The single-copy versus collective comparison has been made on hardware: Peng et al. [20] implement a Fredkin gate and report reduced sample complexity for a hybrid two-copy scheme against single-copy shadows, empirically and at fixed settings, without a law locating where the routes cross.

Statistical-to-bias-floor transitions on hardware. A March 2026 study of shadow tomography on an integrated photonic processor [16] reports reconstruction error following $O(M^{-1/2})$ and then saturating at a hardware-determined floor, a transition the authors term a Hardware Horizon. The structure is adjacent to Section 5 and the two should be distinguished: their transition is between statistical scaling and a bias floor at fixed system size, ours is a change in the statistical exponent itself as a function of system size, occurring before any hardware bias enters. The rolling-exponent diagnostic is likewise not ours — Aasen and Gärttner [21] fit a power-law exponent to adaptive-tomography infidelity in sliding budget windows and report it drifting from $1/N$ toward $1/\sqrt{N}$ — but their migration is driven by detector noise eroding an adaptive advantage, while the one in Section 3.6 is a property of the noiseless estimator's own variance.

Sample-complexity lower bounds for purity. Gong et al. [4] improve the lower bound for purity estimation to $\Omega(\max\{1/\varepsilon^2,\, 2^{n/2}/\varepsilon\})$ for protocols using an identical single-copy projection-valued measurement. Our exact variance is consistent with these bounds and supplies the state-dependent constant they leave open, by evaluating the second moment of [1] over the ensemble.

Collective measurement on noisy hardware. The exponential single-copy versus collective separation is established in [2, 3]; its fate under noise is the subject of [6], the closest theoretical work to ours. For purity *testing* under per-qubit depolarizing noise $D_\lambda(\rho) = (1-\lambda)\rho + \lambda I/2$, their Theorem 2.4 proves a sample-complexity lower bound of

$$\Omega\Big(\min\Big\{\, 2^{n/2},\ \Big(\tfrac{4}{1+3(1-\lambda)^4}\Big)^n \Big\}\Big)\,,$$

writing $b(\lambda) = 4/(1+3(1-\lambda)^4)$ for that base, and it holds against *any* algorithm, including arbitrary adaptive joint POVMs on the $2n$ qubits.

The relationship to our result turns on the task. They test, deciding between hypotheses whose gap shrinks exponentially under noise; we estimate at fixed additive ε, where the bounded collective-test outcome costs $O(1/\varepsilon^2)$ shots regardless of n. Their exponential cost comes from the testing task requiring an exponentially small ε, not from the SWAP test being intrinsically exponential for estimation. We do not evade their bound; we solve a different problem. The two results nonetheless share a mechanism: our Section 4.2 identity, that a per-qubit channel makes the k-copy test return exactly $\mathrm{Tr}(\sigma^k)$ with $\sigma = \mathcal{E}^{\otimes n}(\rho)$, is the finite-$n$ mechanism underneath their asymptotic theorem, since depolarizing drives $\mathrm{Tr}(\sigma^2)$ toward 2^{-n} under both hypotheses and the testing gap closes. Feeding our exact law into their setup at $\lambda = 0.1$, the gap $\Delta(n,\lambda) = \mathrm{Tr}(\sigma^2_{\mathrm{pure}}) - 2^{-n}$ falls from 0.54 at $n = 2$ to 0.21 at $n = 10$, and the implied SWAP-test shot count $1/\Delta^2$ rises from 3.5 to 22 over that window. Their bound is asymptotic with its constant left open, so the comparison available is one of exponential bases rather than counts at any size: the asymptotic base of $1/\Delta^2$, $\big(4/(1+3(1-\lambda)^2)\big)^2$, equals $b(\lambda)$ at $\lambda = 0$ and exceeds it at finite noise, the difference controlled exactly by $3(1-(1-\lambda)^2)^2 \geq 0$; over $n \leq 10$ the large prefactor dominates this small-base exponential, so the growth rate measured on that window sits just below $b(\lambda)$. At our rates and sizes their bound imposes no practical obstruction, because $b \approx 1.35 < \sqrt{2}$ at $\lambda = 0.1$, so the $2^{n/2}$ branch never binds and the cost grows slowly; it becomes an obstruction only at much larger n. Their own framing agrees that the asymptotic limit at fixed noise is not the near-term regime, and that what is needed is a quantitative condition on λ for a two-copy advantage at fixed n. Our crossover criterion is that condition.

NISQ device stability. Dasgupta and Humble [11, 12] quantify the temporal and spatial instability of superconducting devices over 22 months, and related work analyzes calibration drift across devices [13]. Section 6.6 is consistent with their conclusions in the specific setting of collective-measurement protocols and claims no priority over them.

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8. LIMITATIONS

The closed forms are $k = 2$ only, and the prefactor is asymptotic. The second-order projection variances have closed forms for the ensembles considered here, and $M^*(n)$ is analytic for them (Section 3.5). Three limits apply. The evaluation uses the Haar moments of a specific ensemble, so for an arbitrary state the identity of [1] must be summed over that state's Pauli spectrum. At $k \geq 3$ we did not obtain closed forms and the coefficients are estimated numerically. And the asymptotic base and prefactor, though exact, are approached slowly: at the sizes reached here $M^*(n)/5.6^n$ is still some fifteen percent below its limit.

The $k = 3$ validation is 6 of 7. The single miss sits on the two-sigma boundary and is sensitive to the random seed. We report it rather than round it to 7 of 7.

The noise model is a channel abstraction. Global depolarizing and independent per-qubit channels are idealizations. The bias laws are exact given the channel, and they say nothing about whether the channel describes any particular device. Section 6 documents exactly where a real device departs from them.

The crossover was not demonstrated on hardware. It sits at $n \approx 5$ to 8 under realistic noise, and cross-session variation dominates at those sizes on the hardware available to us. Section 6 is a test of the model, not a demonstration of the advantage.

The hardware single-copy baseline at $n \geq 3$ is computed, not measured, for the economic reason given in Section 6.6. The one point we could afford to measure, the $n = 2$ anchor, missed its prediction by 5.6 predicted standard deviations (Section 6.4). The computed points are therefore model output on a device that our own anchor shows the model does not currently track, and nothing in Section 6 should be read as a hardware measurement of the single-copy route.

The execution path on the Public Tier is not independently verifiable by us. Jobs were routed through a decentralized network with validator spot-checking rather than direct vendor access (Section 6.1). We cannot exclude this as a contributor to the session-to-session variability of Section 6.4. It does not affect the readout and gate characterizations, which are internally consistent, but a replication with direct vendor access would be worth running.

Statistical caveats. The out-of-ensemble RMSE validation carries a median relative deviation of 6.7% between predicted and measured values (Fig. 4). We investigated this and found it to be finite-trial noise on both sides rather than a systematic error: the estimator is approximately Gaussian at the budgets used, the projection variances are converged to better than 1%, and the measured RMSE converges to the predicted value to within $\pm 0.3\%$ as the trial count increases. The 6.7% figure matches the intrinsic RMSE noise at the trial counts used, and the signed deviation is $+3.4\%$, which is within noise once cell correlations are accounted for. The predicted crossover n^*_{pred} is reported as an integer throughout, and we do not propagate the projection-variance estimation uncertainty onto it. Resampling the projection variances by their committed relative spread shifts the predicted crossover for 5 of the 60 noisy-pure cells (mean per-cell probability 1.7%), always by exactly one qubit and never by more.

The readout sensitivity check is a simulation. It uses a single readout model, the measured per-qubit rates of Section 6.3 and a uniform 6.5% comparator, with independent and time-invariant readout and no correlated or drifting component; and the collective route's gate, depth, and routing overhead is not folded into its RMSE. It bounds the direction of the crossover shift, not its location on hardware.

9. CONCLUSION

We began with a number: at ten qubits, on noisy-pure states at a fixed budget, single-copy estimation of purity returns an error fifteen times the quantity being estimated. That number is not an accident of implementation, and this paper explains where it comes from.

That the effective budget-scaling exponent is not a constant is established [19, 21]; where it changes is not, because separate upper bounds on ζ_1 and ζ_2 constrain their ratio not at all. Evaluating them fixes the threshold, and with it the exponent over the budget window an experiment can reach. The asymptotic scaling at fixed system size is unchanged and remains the square-root law.

Paired with two exact bias laws for the collective route, distinguished by the geometry of the noise channel rather than by its rate, the variance law predicts without free parameters the size at which collective measurement attains the lower RMSE at equal copy budget. It places 82 of the 83 simulated crossovers within one qubit.

The practical guidance follows. For estimating purity of noisy-pure states under per-qubit amplitude-damping or dephasing noise at the rates and budgets tested here, single-copy classical shadows stop being the lower-RMSE route beyond roughly six to seven qubits; the higher moments hold out to about eight for the idealized bounded-outcome test, whose $k \geq 3$ realization costs about an order of magnitude more two-qubit gates (Section 2.3), and under global depolarizing noise the crossover falls earlier still. In every case the law says which side of the boundary a given experiment is on. The entangling overhead that has made the field cautious is, for the structured $n = 2$ circuits on matched topology we ran and under the assumed median CZ error, a minor contribution: the test adds two two-qubit gates of measurement overhead on top of the two that prepare the copies, with no routing, and under that error model the four together account for about a seventh of the observed error at the general-availability median, or close to a fifth at the lower median the vendor has since published, though edge-specific CZ error on these circuits was not measured.

What blocks the demonstration is neither variance nor, under the model we assume, gate error. It is readout fidelity, three times the value we assumed and strongly correlated with neighboring excitations, and above all the session-to-session variation of the execution path, whose mechanism we cannot isolate. Cross-session drift on byte-identical circuits exceeded within-session drift by a factor of twenty and exceeded every modeled error source, and could not be attributed to any of seven candidate mechanisms, four excluded outright and three bounded well below the deficit. That is more specific than a call for better gates: readout fidelity and run-to-run stability, neither of which the published specification for this backend gives, are what stand between current devices and the advantage the criterion places near six qubits.

Two questions follow. Whether the same variance law, applied to the partial-transpose moments, gives an exact sample-complexity account of entanglement negativity estimation, which is the application that motivated us. And whether the drift we measured is a property of this device, this platform, or this class of hardware, which requires a multi-session, multi-device study we did not have the access to run.

ACKNOWLEDGEMENTS

Quantum hardware access was provided by the Open Quantum platform operated by Quantum Rings Inc. [9], on the Public Tier, whose license conditions require citation of the platform paper and contribution of anonymized aggregated circuit data to a common repository. All hardware experiments ran on the backend reported by the platform as Rigetti Computing's Cepheus-1-108Q processor. [Mentor acknowledgement.]

DATA AND CODE AVAILABILITY

All code, raw measurement counts, and analysis are available at `https://github.com/alex-messerlian/shadow-moment-crossover`. Raw hardware counts were committed to version control before any analysis was performed, and every locked prediction was committed before the corresponding measurement was submitted.

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Appendix A: The exact fourth-moment U-statistic

The full U-statistic for $\mathrm{Tr}(\rho^4)$ requires the sum over all distinct ordered 4-tuples of snapshots,

$$S = \sum_{(i,j,k,l)\ \text{pairwise distinct}} \mathrm{Tr}(G_i G_j G_k G_l).$$

Because the snapshots do not commute, this does not reduce to matrix power sums. It is obtained by Möbius inversion over the partition lattice of the four cyclic slots.

For each set partition P of $\{0,1,2,3\}$, let $T(P)$ be the sum of $\mathrm{Tr}(G_{a_0} G_{a_1} G_{a_2} G_{a_3})$ over all index assignments in which slots in the same block share an index and different blocks are unconstrained. The Möbius function of the partition lattice is

$$\mu(P) = \prod_{b \in P} (-1)^{|b|-1} (|b|-1)!,$$

and the all-distinct sum is

$$S = \sum_P \mu(P)\, T(P),$$

summed over the 15 set partitions of four elements.

Most $T(P)$ reduce to matrix power sums built from $S_1 = \sum_i G_i$, $P_2 = \sum_i G_i^2$, $P_3 = \sum_i G_i^3$, and $P_4 = \sum_i G_i^4$. The exception is the alternating partition $\{0,2\}\{1,3\}$, which requires

$$A = \sum_{i,j} \mathrm{Tr}(G_i G_j G_i G_j),$$

and this has no power-sum representation. It is computed exactly by a tensor contraction. Define the $d^2 \times d^2$ matrix

$$X_{(a,b),(c,d)} = \sum_i (G_i)_{ab} (G_i)_{cd} = \sum_i \mathrm{vec}(G_i)\, \mathrm{vec}(G_i)^\top,$$

reshape it to X_{abcd}, and contract

$$A = \sum_{a,b,c,d} X_{abcd}\, X_{bcda}.$$

Forming X costs $\Theta(Md^4)$ in time, since each of the M rank-one updates touches all d^4 entries, and $\Theta(d^4)$ in memory; the two contractions cost $\Theta(d^4)$. The construction as written is therefore $\Theta(Md^4)$. The estimator used for the results reported here never forms X: it evaluates the alternating term through the per-qubit factorization $\mathrm{Tr}(G_iG_jG_iG_j) = \prod_q \mathrm{Tr}(G_i^{(q)}G_j^{(q)}G_i^{(q)}G_j^{(q)})$ as an $M \times M$ contraction, which costs $O(M^2n + Mnd^2)$ and forms no d^4 tensor. Both routes agree, and the complete $k = 4$ estimator matches brute-force enumeration over all distinct 4-tuples to $\lesssim 10^{-13}$.

Appendix B: The destructive SWAP test

Two copies of an n-qubit state occupy qubits $0 \ldots n-1$ (copy A) and $n \ldots 2n-1$ (copy B). For each $i \in \{0, \ldots, n-1\}$, apply a CNOT with control i and target $i+n$, then a Hadamard on qubit i. Measure all $2n$ qubits. For each measured bitstring, let a_i and b_i be the outcomes on qubits i and $i+n$. The purity is

$$\mathrm{Tr}(\rho^2) = \sum_{\text{bitstrings}} (-1)^{\#\{i\,:\,a_i=b_i=1\}}\, P(\text{bitstring}).$$

The circuit uses exactly n two-qubit gates, no ancilla, and has depth two after state preparation. The sign rule is invariant under bitstring reversal, so the endianness convention of the backend does not affect the result. We verified the rule against exact simulation for pure and mixed states to 10^{-16}.

Appendix C: Reproducibility and statistics

This appendix records the sampling design behind every headline number, so that the agreement between prediction and measurement can be judged against how the two were generated. All counts are read from the committed generating code; the sizes and seeds below are those actually used.

State and snapshot counts. The projection variances ζ_c that feed the theory curves are estimated from 4 independently drawn states per (n, k), each from 120,000 shadow snapshots (`results/theory_zetas.json`: `n_states` = 4, `n_samples` = 120,000). At $k = 2$ these coefficients are now computed in closed form (Section 3.5) and carry no sampling error; the sampled values are retained in the repository, and substituting them changes no crossover prediction and no α verdict. The measured α points and the noisy-pure crossover cells are averaged over 48 independently drawn states per size, 8 Monte Carlo trials each (`run_budget_sweep.py`: `N_STATES` = 48, `N_TRIALS` = 8). The held-out stress ensembles use 4 states for the variance estimate and 24 for the measurement on the random families (Haar-pure, low-rank), and a single fixed state with 36 trials for the deterministic GHZ-noisy family (`run_stress_test.py`); their variance estimates use 60,000 snapshots. All states are depolarized Haar-random pure states at $q = 0.1$, except the low-rank and GHZ-noisy families defined in Section 4.3.

Train/test separation. The prediction and the measurement it is compared against do not share snapshots. The ζ_c are estimated from one pseudorandom substream (seeded $[\,0, n, s, k, 3\,]$) and the measured RMSE from a disjoint one (seeded $[\,0, n, s, 1, k, t\,]$), so the shot noise in the theory curve and the shot noise in the measured points are independent by construction. The theory curve is never fit to the measured α: it is the exact-variance RMSE evaluated at the estimated ζ_c and fit over the budget grid on its own (`figures/data.py`: the components are read from `theory_zetas.json` and "not re-estimated"). The state draws do overlap: the four states used for ζ_c are, at this seed, the first four of the forty-eight used for the measured points, since both draw ρ from `noisy_pure`$(n, 0.1, \mathrm{rng}[0, n, s, 0])$. The agreement is therefore not built in through shared shots, though it is not a fully held-out state set either.

Bootstrap resampling units. The α standard errors resample the state as the independent unit (2000 resamples, each carrying its whole correlated budget vector; `fit_budget_exponent_bootstrap`), which is the honest unit because the nested budgets share a draw. The hardware confidence intervals resample differently and less cleanly: the single-copy anchor resamples the 9000 snapshots (300 resamples, "within the fixed 15 bases"), and the collective and Bell-SWAP intervals resample per-shot ± 1 values (4000 and 5000 resamples). Shots within one circuit and one calibration are not independent draws of the estimator, so these hardware intervals capture shot noise conditional on the realized circuit and calibration, not the full run-to-run variability; they should be read as lower bounds on the true uncertainty.

The α regression. Both the measured and the theory α are the negative ordinary-least-squares slope of $\log \mathrm{RMSE}$ against $\log M$ over the same per-cell budget grid ($k = 2$: 2000–128,000, capped to 2000–32,000 for $n \geq 7$; $k = 3$: 2000–32,000; $k = 4$: 500–8000), with

$\mathrm{RMSE}(M) = \sqrt{\mathrm{MSE}}$ averaged over states. The two differ only in their input — measured MSE versus the exact-variance RMSE — and are otherwise the identical procedure.

Crossover definition. The crossover n^* is the smallest size from which the single-copy RMSE exceeds the collective RMSE and stays above it for the remainder of the swept range (the sustained crossover of `predict_crossover`). It is an integer from the swept sizes; no interpolation is used, and a cell whose curves do not cross inside the tested range is censored (returns none) and excluded from the aggregate rather than counted as a hit or a miss.

Fit uncertainty. The exponential-fit residuals in Section 3.6 are positive at both ends of the range ($+0.20$ at $n = 2$, $+0.13$ at $n = 9$ in $\log M^*$) and negative through the middle, the signature of mild upward curvature; the fitted base accordingly rises with the window, 5.14 over $n = 2$–7, 5.34 over 2–9, 5.60 over 4–9. Refitting with each n dropped in turn moves the base within $[5.25, 5.49]$, with the low-n point exerting the most leverage (dropping $n = 2$ raises it to 5.49). The base is a finite-size fit over the sizes reached, not an estimate of an asymptotic rate.

The full crossover table. The 83 resolved cells break down as: 68 noisy-pure cells drawn from two independent sweeps (`budget_scaling.json` and `moment_sweep_corrected.json`), which comprise 54 distinct $(k, \mathrm{channel}, \mathrm{rate}, \mathrm{budget})$ configurations with 14 evaluated in both sweeps — and all 14 replications agree on both the measured and the predicted n^* — plus 15 cells across the three held-out ensembles. Of the 123 cells swept in total, these resolved a sustained crossover within the tested range; the remaining 40 — 28 noisy-pure and 12 held-out — were censored because their single-copy and collective curves do not cross inside the swept range, and are excluded from the aggregate rather than scored as a hit or a miss. Treating "no crossover in the swept range" as a valid prediction and scoring all swept cells, the criterion is exact on 109 and within one qubit on 118, or 88.6% and 95.9%; the censored cells are not random, concentrating at the largest budgets, where the crossover is pushed past the swept range, and in the Haar-pure ensemble. Among the censored cells the criterion agrees on no crossover in 36 and misses in 4, so the censoring discards mostly correct no-crossover predictions but also removes four misses that would lower the within-one-qubit figure to 95.9%. The supplementary material lists all 83 individually, and the same cells are committed as `budget_scaling.json`, `moment_sweep_corrected.json`, and `stress_test.json` in the repository. One cell misses by more than one qubit (noisy-pure, $k = 4$, predicted 4 against measured 2), giving $82/83 = 98.8\%$ within one qubit and $73/83 = 88.0\%$ exact. Stratified by the measured crossover, all 52 cells with $n^*_{\text{meas}} \geq 3$ fall within one qubit and 43 are exact; the one miss beyond a qubit sits in the 31-cell $n^*_{\text{meas}} = 2$ group. Counted as configurations rather than evaluations, the 83 cells are 69 independent settings — 54 distinct noisy-pure plus 15 held-out — with 14 noisy-pure replications.